AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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Total Marks: 100

Examination Paper

Question Q1 (20 marks)

a.i) List the three types of integrity constraints discussed in today's lecture. *(4)*

a.ii) Explain the concept of referential integrity and its importance in a relational database schema. *(4)*

a.iii) Describe the process of normalizing a relation to achieve Boyce-Codd Normal Form (BCNF). *(5)*

b) Design a relational schema for a university database that includes the following entities: Students, Courses, Professors, and Departments. The schema should include the following attributes: Students - StudentId (primary key), Name, Email, CourseId (foreign key to Courses), ProfessorId (foreign key to Professors); Courses - CourseId (primary key), CourseName, Credits; Professors - ProfessorId (primary key), Name, DepartmentId (foreign key to Departments); Departments - DepartmentId (primary key), DepartmentName. *(7)*

Question Q2 (20 marks)

a) Draw an ER diagram for the scenario below. A Library is organized into several sections such as fiction, children and technology. Each section has a name and a number(unique) and its headed by a head librarian. Each book belong to a section and has a title, authors, ISBN, year and a publisher. A book may have several copies. Each copy is identified by an access number. For each copy borrowed, current borrower and due date should be tracked. Members have a membership number, an address and a phone number. Members can borrow 5 books and could put hold request on a book. Librarian has a name, id number, phone and an address. *(4)*

b) List three properties of the Boyce-Codd Normal Form (BCNF). *(6)*

c) Define functional dependency and explain its importance in normalization. Provide an example to illustrate your answer. *(10)*

Question Q3 (22 marks)

a)) List the three main components of an ER model. *(2)*

b)) Explain the concept of fan trap in ER modeling and provide an example. *(2)*

c)) Define aggregation relationship and explain its difference with ternary relationship. *(2)*

d)) Explain the concept of overlapping constraint and provide an example. *(3)*

e).i)) List the five types of integrity constraints in database management systems. *(2)*

e).ii)) Explain the concept of relational model and its components. *(3)*

e).iii)) Define domain constraint and explain its purpose in database management systems. *(2)*

e).iv)) Explain the concept of referential integrity constraints and provide an example. *(3)*

e).v)) List the benefits of using a database management system over file processing systems. *(3)*

Question Q4 (20 marks)

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a) Explain the concept of Aggregation in ER modeling. (2 marks) *(2)*

b) List three types of integrity constraints used in relational database design. (3 marks) *(3)*

c.i) Draw an ER diagram for the following scenario: Students contain an id (unique), name and an address. There are academic semesters containing an semester id (unique), semester and year. There are courses offered during academic semesters. A course has a number (unique), name and credits. Students make payments. A payment has receipt number (unique), amount and date. Payments can be classified into Tuition (semester payment), Examination and other (Library fine, Printouts). A Tuition payment is made for an academic semester For other payments description should be stored. Students register for courses offered during a particular semester. The registered date must be stored in the database. (4 marks) *(4)*

c.ii) Explain the concept of Referential Integrity Constraint and its importance in relational database design. (2 marks) *(2)*

d.i) List two common types of relationships used in ER modeling: Ternary and Aggregation. (2 marks) *(2)*

d.ii) Construct an ER diagram for the following scenario: A library is organized into several sections such as fiction, children and technology. Each section has a name and a number(unique) and its headed by a head librarian. Each book belongs to a section and has a title, authors, ISBN, year and a publisher. Each copy is identified by an access number. For each copy borrowed, current borrower and due date should be tracked.

Members have a membership number, an address and a phone number. Members can borrow 5 books and could put hold request on a book. Librarian has a name, id number, phone and an address. (7 marks) *(7)*

Question Q5 (18 marks)

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a)) List at least three properties of a relation that is in Second Normal Form (2NF). *(5)*

b)) Define functional dependency and explain its importance in database normalization. *(4)*

c)) Consider a relation R (A, B, C, D) with the following set of functional dependencies over R: F = { A -> B, B -> C, B -> D}. Compute the keys for relation R. *(9)*